```
Iteration 1: INITIAL STATE
State: [7, 1, 5, 3, 1, 1, 2, 4]
Cost: 8
  0 1 2 3 4 5 6 7
0 . . . . . . . .
1 . Q . . Q Q . .
2 . . . . . . Q .
3 . . . Q . . . .
4 . . . . . . . Q
5 . . Q . . . . .
6 . . . . . . . .
7 Q . . . . . . .

Iteration 2: INTERMEDIATE STATE
State: [7, 1, 1, 3, 1, 1, 2, 4]
Cost: 10
  0 1 2 3 4 5 6 7
0 . . . . . . . .
1 . Q Q . Q Q . .
2 . . . . . . Q .
3 . . . Q . . . .
4 . . . . . . . Q
5 . . . . . . . .
6 . . . . . . . .
7 Q . . . . . . .

Iteration 3: INTERMEDIATE STATE
State: [7, 6, 1, 3, 1, 1, 2, 4]
Cost: 7
  0 1 2 3 4 5 6 7
0 . . . . . . . .
1 . . Q . Q Q . .
2 . . . . . . Q .
3 . . . Q . . . .
4 . . . . . . . Q
5 . . . . . . . .
6 . Q . . . . . .
7 Q . . . . . . .

Iteration 4: INTERMEDIATE STATE
State: [4, 6, 1, 3, 1, 1, 2, 4]
Cost: 7
  0 1 2 3 4 5 6 7
0 . . . . . . . .
1 . . Q . Q Q . .
2 . . . . . . Q .
3 . . . Q . . . .
4 Q . . . . . . Q
5 . . . . . . . .
6 . Q . . . . . .
7 . . . . . . . .

Iteration 5: INTERMEDIATE STATE
State: [4, 7, 1, 3, 1, 1, 2, 4]
Cost: 8
  0 1 2 3 4 5 6 7
0 . . . . . . . .
1 . . Q . Q Q . .
2 . . . . . . Q .
3 . . . Q . . . .
4 Q . . . . . . Q
5 . . . . . . . .
```

```
Iteration 6: INTERMEDIATE STATE
State: [4, 7, 1, 3, 2, 1, 2, 4]
Cost: 8
  0 1 2 3 4 5 6 7
0 . . . . . . . .
1 . . Q . . Q . .
2 . . . . Q . Q .
3 . . . Q . . . .
4 Q . . . . . . Q
5 . . . . . . . .
6 . . . . . . . .
7 . Q . . . . . .

Iteration 7: INTERMEDIATE STATE
State: [5, 7, 1, 3, 2, 1, 2, 4]
Cost: 7
  0 1 2 3 4 5 6 7
0 . . . . . . . .
1 . . Q . . Q . .
2 . . . . Q . Q .
3 . . . Q . . . .
4 . . . . . . . Q
5 Q . . . . . . .
6 . . . . . . . .
7 . Q . . . . . .

Iteration 8: INTERMEDIATE STATE
State: [5, 7, 1, 3, 2, 1, 4, 4]
Cost: 6
  0 1 2 3 4 5 6 7
0 . . . . . . . .
1 . . Q . . Q . .
2 . . . . Q . . .
3 . . . Q . . . .
4 . . . . . Q Q .
5 Q . . . . . . .
6 . . . . . . . .
7 . Q . . . . . .

Iteration 9: INTERMEDIATE STATE
State: [5, 7, 1, 3, 2, 0, 4, 4]
Cost: 4
  0 1 2 3 4 5 6 7
0 . . . . . Q . .
1 . . Q . . . . .
2 . . . . Q . . .
3 . . . Q . . . .
4 . . . . . Q Q .
5 Q . . . . . . .
6 . . . . . . . .
7 . Q . . . . . .

Iteration 10: INTERMEDIATE STATE
State: [5, 7, 1, 3, 2, 0, 6, 4]
Cost: 3
  0 1 2 3 4 5 6 7
0 . . . . . Q . .
1 . . Q . . . . .
2 . . . . Q . . .
3 . . . Q . . . .
```

```
6 . . . . . Q .
7 . Q . . . . .

Iteration 11: INTERMEDIATE STATE
State: [5, 1, 1, 3, 2, 0, 6, 4]
Cost: 6
  0 1 2 3 4 5 6 7
0 . . . . . Q . .
1 . Q Q . . . . .
2 . . . . Q . . .
3 . . . Q . . . .
4 . . . . . . . Q
5 Q . . . . . . .
6 . . . . . Q . .
7 . . . . . . . .

Iteration 12: INTERMEDIATE STATE
State: [5, 1, 1, 3, 2, 0, 6, 0]
Cost: 7
  0 1 2 3 4 5 6 7
0 . . . . . Q . Q
1 . Q Q . . . . .
2 . . . . Q . . .
3 . . . Q . . . .
4 . . . . . . . .
5 Q . . . . . . .
6 . . . . . Q . .
7 . . . . . . . .

Iteration 13: INTERMEDIATE STATE
State: [5, 2, 1, 3, 2, 0, 6, 0]
Cost: 6
  0 1 2 3 4 5 6 7
0 . . . . . Q . Q
1 . . Q . . . . .
2 . Q . . Q . . .
3 . . . Q . . . .
4 . . . . . . . .
5 Q . . . . . . .
6 . . . . . Q . .
7 . . . . . . . .

Iteration 14: INTERMEDIATE STATE
State: [5, 2, 1, 3, 2, 4, 6, 0]
Cost: 5
  0 1 2 3 4 5 6 7
0 . . . . . . . Q
1 . . Q . . . . .
2 . Q . . Q . . .
3 . . . Q . . . .
4 . . . . . Q . .
5 Q . . . . . . .
6 . . . . . Q . .
7 . . . . . . . .

Iteration 15: INTERMEDIATE STATE
State: [7, 2, 1, 3, 2, 4, 6, 0]
Cost: 6
  0 1 2 3 4 5 6 7
0 . . . . . . . Q
1 . . Q . . . . .
2 . Q . . Q . . .
```

```
4 . . . . . Q . .
5 . . . . . . . .
6 . . . . . Q .
7 Q . . . . . .

Iteration 16: INTERMEDIATE STATE
State: [7, 2, 1, 1, 2, 4, 6, 0]
Cost: 6
  0 1 2 3 4 5 6 7
0 . . . . . . . Q
1 . . Q Q . . . .
2 . Q . . Q . . .
3 . . . . . . . .
4 . . . . . Q . .
5 . . . . . . . .
6 . . . . . Q .
7 Q . . . . . .

Iteration 17: INTERMEDIATE STATE
State: [7, 2, 3, 1, 2, 4, 6, 0]
Cost: 4
  0 1 2 3 4 5 6 7
0 . . . . . . . Q
1 . . . Q . . . .
2 . Q . . Q . . .
3 . . Q . . . . .
4 . . . . . Q . .
5 . . . . . . . .
6 . . . . . Q .
7 Q . . . . . .

Iteration 18: INTERMEDIATE STATE
State: [7, 2, 0, 1, 2, 4, 6, 0]
Cost: 6
  0 1 2 3 4 5 6 7
0 . . Q . . . . Q
1 . . . Q . . . .
2 . Q . . Q . . .
3 . . . . . . . .
4 . . . . . Q . .
5 . . . . . . . .
6 . . . . . Q .
7 Q . . . . . .

Iteration 19: INTERMEDIATE STATE
State: [7, 2, 0, 1, 3, 4, 6, 0]
Cost: 6
  0 1 2 3 4 5 6 7
0 . . Q . . . . Q
1 . . . Q . . . .
2 . Q . . . . . .
3 . . . . Q . . .
4 . . . . . Q . .
5 . . . . . . . .
6 . . . . . Q .
7 Q . . . . . .

Iteration 20: INTERMEDIATE STATE
State: [7, 2, 0, 7, 3, 4, 6, 0]
Cost: 6
  0 1 2 3 4 5 6 7
0 . . Q . . . . Q
```

```
Iteration 21: INTERMEDIATE STATE
State: [7, 2, 0, 7, 6, 4, 6, 0]
Cost: 5
  0 1 2 3 4 5 6 7
0 . . Q . . . . Q
1 . . . . . . . .
2 . Q . . . . . .
3 . . . . . . . .
4 . . . . . Q . .
5 . . . . . . . .
6 . . . . Q . Q .
7 Q . . Q . . . .

Iteration 22: INTERMEDIATE STATE
State: [7, 2, 0, 0, 6, 4, 6, 0]
Cost: 6
  0 1 2 3 4 5 6 7
0 . . Q Q . . . Q
1 . . . . . . . .
2 . Q . . . . . .
3 . . . . . . . .
4 . . . . . Q . .
5 . . . . . . . .
6 . . . . Q . Q .
7 Q . . . . . . .

Iteration 23: INTERMEDIATE STATE
State: [7, 2, 0, 0, 6, 3, 6, 0]
Cost: 7
  0 1 2 3 4 5 6 7
0 . . Q Q . . . Q
1 . . . . . . . .
2 . Q . . . . . .
3 . . . . . Q . .
4 . . . . . . . .
5 . . . . . . . .
6 . . . . Q . Q .
7 Q . . . . . . .

Iteration 24: INTERMEDIATE STATE
State: [7, 2, 0, 6, 6, 3, 6, 0]
Cost: 6
  0 1 2 3 4 5 6 7
0 . . Q . . . . Q
1 . . . . . . . .
2 . Q . . . . . .
3 . . . . . Q . .
4 . . . . . . . .
5 . . . . . . . .
6 . . . Q Q . Q .
7 Q . . . . . . .

Iteration 25: INTERMEDIATE STATE
State: [7, 2, 0, 6, 6, 3, 3, 0]
Cost: 6
  0 1 2 3 4 5 6 7
0 . . Q . . . . Q
1 . . . . . . . .
2 . Q . . . . . .
3 . . . . . Q Q .
4 . . . . . . . .
```

```
1 . . . . . .
2 . Q . . . . .
3 . . . . . Q .
4 . . . . . .
5 . . . . . .
6 . . . Q Q . . .
7 Q . . . . Q . Q

Iteration 29: INTERMEDIATE STATE
State: [7, 2, 0, 6, 6, 4, 3, 7]
Cost: 5
  0 1 2 3 4 5 6 7
0 . . Q . . . . .
1 . . . . . .
2 . Q . . . . .
3 . . . . . Q .
4 . . . . . Q . .
5 . . . . . .
6 . . . Q Q . . .
7 Q . . . . . Q

Iteration 30: INTERMEDIATE STATE
State: [7, 2, 0, 6, 6, 4, 3, 1]
Cost: 4
  0 1 2 3 4 5 6 7
0 . . Q . . . . .
1 . . . . . Q
2 . Q . . . . .
3 . . . . . Q .
4 . . . . . Q . .
5 . . . . . .
6 . . . Q Q . . .
7 Q . . . . . .

Iteration 31: INTERMEDIATE STATE
State: [7, 2, 0, 6, 6, 4, 7, 1]
Cost: 4
  0 1 2 3 4 5 6 7
0 . . Q . . . . .
1 . . . . . Q
2 . Q . . . . .
3 . . . . . .
4 . . . . . Q . .
5 . . . . . .
6 . . . Q Q . . .
7 Q . . . . . Q .

Iteration 32: FINAL GOAL STATE
State: [0, 4, 7, 5, 2, 6, 1, 3]
Cost: 0
  0 1 2 3 4 5 6 7
0 Q . . . . . .
1 . . . . . Q .
2 . . . . Q . . .
3 . . . . . Q
4 . Q . . . . .
5 . . . Q . . . .
6 . . . . . Q . .
7 . . Q . . . . .

Goal state reached!
All 8 queens are safe!
```